

Eco 202 - Fall 2026

Class 03

$$s_x^2 = \frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})^2 \quad ; \quad \bar{x} = \frac{1}{n} \sum_{i=1}^n x_i$$

$$y_i = a + bx_i$$

$$s_y^2 = \frac{1}{n-1} \sum_{i=1}^n (y_i - \bar{y})^2 \quad ; \quad \bar{y} = \frac{1}{n} \sum_{i=1}^n y_i$$

$$= \frac{1}{n-1} \sum_{i=1}^n \left((a + bx_i) - \frac{1}{n} \sum_{i=1}^n (a + bx_i) \right)^2$$

$$= \frac{1}{n-1} \sum_{i=1}^n \left(a + bx_i - \underbrace{\frac{1}{n} \sum_{i=1}^n a}_{a} - b \frac{1}{n} \sum_{i=1}^n x_i \right)^2$$

$$= \frac{1}{n-1} \sum_{i=1}^n \left(b(x_i - \bar{x}) \right)^2$$

$$= b^2 \cdot s_x^2$$

$$z_i = \frac{x_i - \bar{x}}{\sqrt{s_x^2}} = \frac{x_i - \bar{x}}{s_x}$$

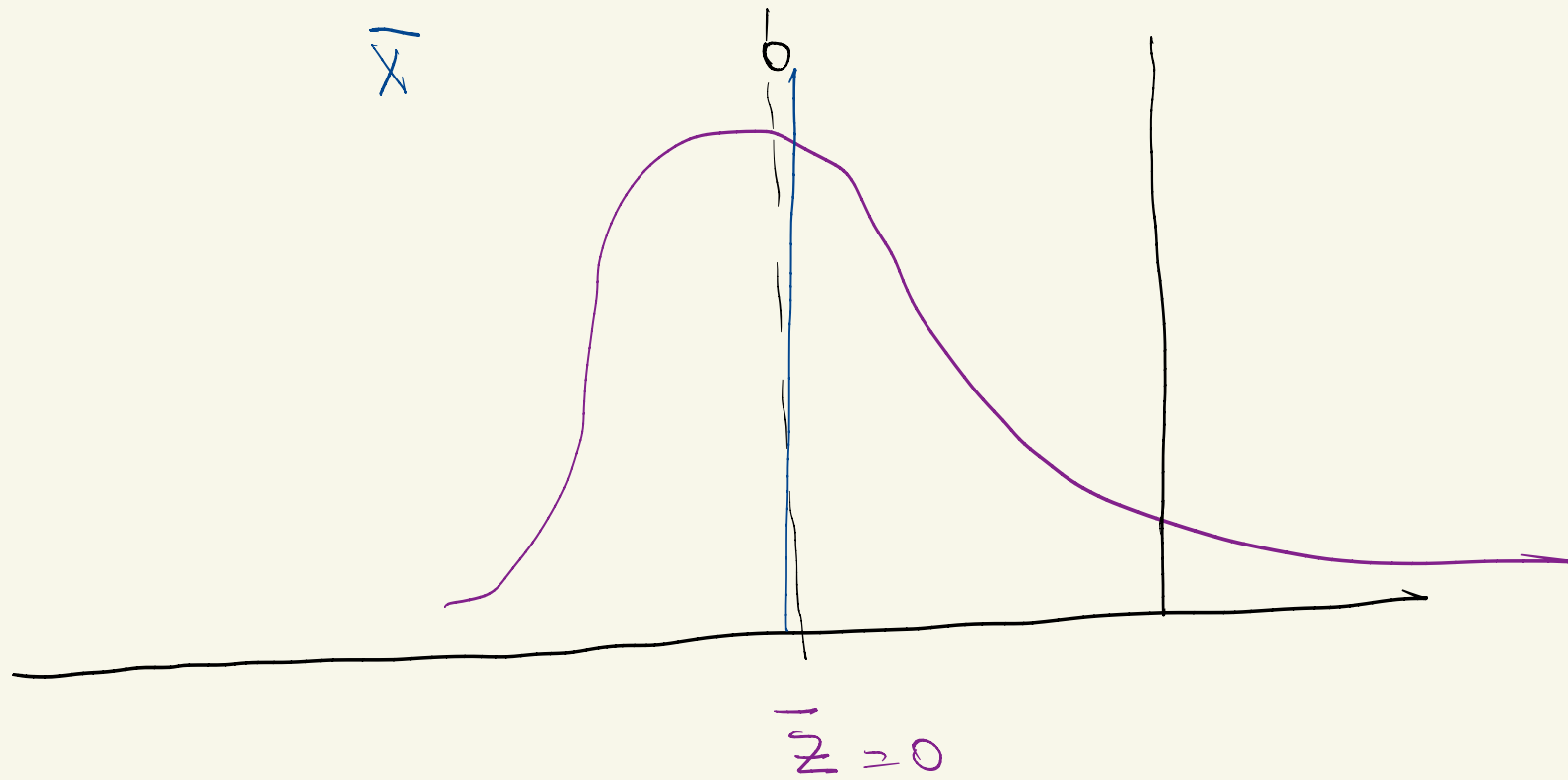
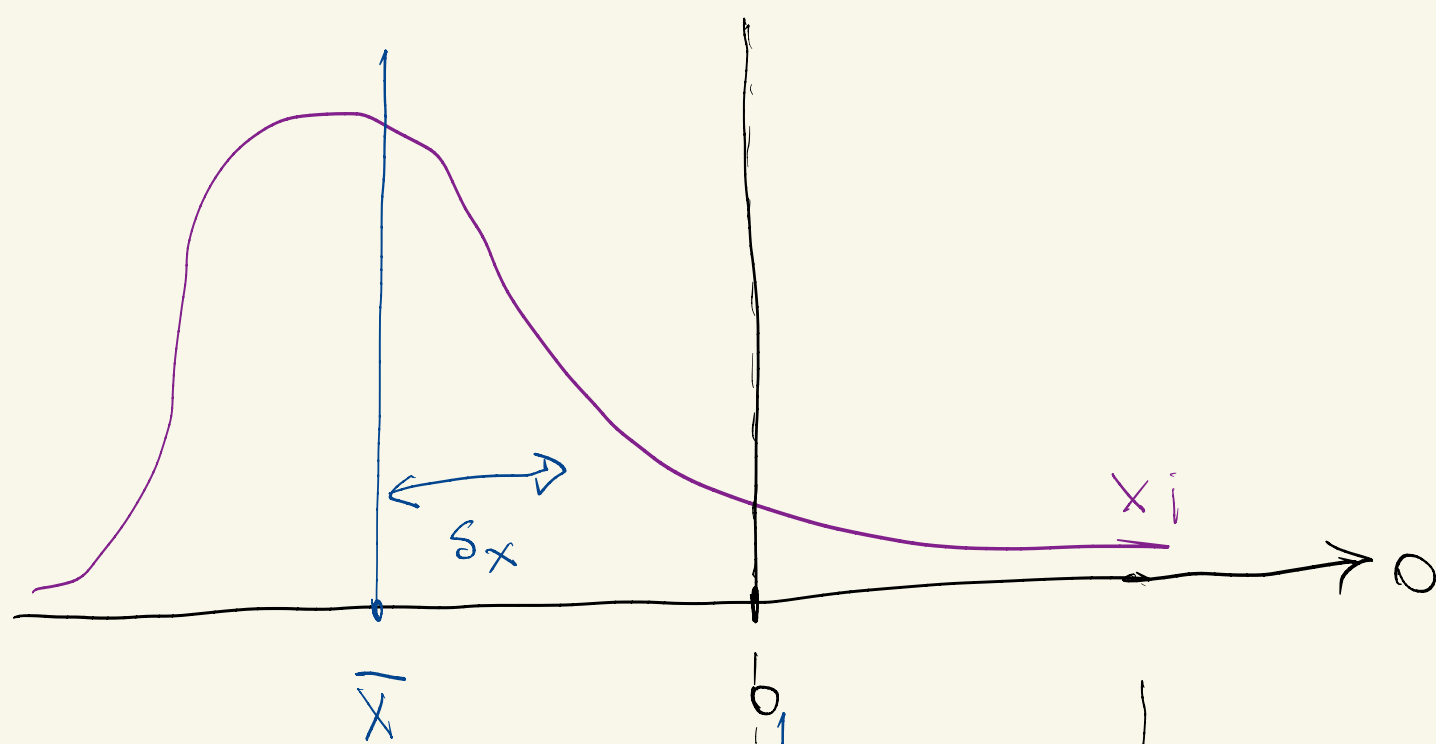
$$\frac{1}{n} \sum_{i=1}^n z_i = \frac{1}{n} \sum_{i=1}^n \left(\frac{x_i - \bar{x}}{s_x} \right)$$

$$= \frac{1}{s_x} \cdot \frac{1}{n} \sum_{i=1}^n (x_i - \bar{x})$$

$$= \frac{1}{s_x} \left(\underbrace{\frac{1}{n} \sum_{i=1}^n x_i}_{\bar{x}} - \underbrace{\frac{1}{n} \sum_{i=1}^n \bar{x}}_{\bar{x}} \right)$$

= 0

□



$$z_i = \frac{x_i - \bar{x}}{s_x} = b \cdot y_i \Rightarrow s_z^2 = b^2 \cdot s_y^2$$

$$= \underbrace{\frac{1}{s_x}}_b \cdot \underbrace{(x_i - \bar{x})}_{y_i}$$

$$s_z^2 = \frac{1}{n-1} \sum_{i=1}^n (z_i - \bar{z})^2$$

$$= \frac{1}{b^2} \cdot s_y^2 = \frac{1}{s_x^2} \cdot \frac{1}{n-1} \sum_{i=1}^n (y_i - \bar{y})^2$$

$$= \frac{1}{s_x^2} \cdot s_x^2 = 1$$

